

Python Coding Challenge

Instructions:

- 1 Attempt all questions.
 - 2 All solutions must be uploaded in a folder named as "**Coding_Challenge_22nd_May_2026**" in your GitHub repository.
 - 3 Solutions must be uploaded one by one with proper commit messages.
 - 4 Each solution must contain proper comments as well as sample output.
 - 5 Identifiers (variables, functions, classes) must be meaningful and professional.
 - 6 Maintain clean code formatting and readability.
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Q1. A grocery shop owner wants to calculate the final bill after applying discounts based on the purchase amount. Write a program using if...elif...else to display the final amount.

Q2. A company wants to identify whether an employee is eligible for a bonus based on attendance percentage and performance rating.

Q3. Create a program to check whether a given username and password match predefined credentials.

Q4. A traffic monitoring system needs to determine whether a vehicle exceeded the speed limit.

Q5. A bank wants to calculate simple interest for customers based on principal, rate, and time entered by the user.

Q6. A movie theatre wants to assign ticket pricing based on age categories such as child, adult, and senior citizen.

Q7. Write a program to find the largest among three entered sales figures.

Q8. A school wants to calculate grades for students based on marks obtained in five subjects.

Q9. Create a login attempt system that allows the user a maximum of 3 attempts using a while loop.

Q10. A food delivery app wants to display menu items repeatedly until the user selects Exit.

Q11. Write a program to print all even numbers between 1 and 100 using a for loop.

Q12. A library system wants to count how many overdue books exist in a list of borrowed books.

Q13. Create a function that calculates electricity bills based on unit consumption.

Q14. A company wants a reusable function to calculate annual salary from monthly salary.

Q15. Write a function to check whether a given string is a palindrome.

- Q16.** A teacher wants to store marks of students in a list and display the average marks.
- Q17.** Create a program to remove duplicate product IDs from a list using sets.
- Q18.** A travel company wants to store city names visited by a customer in a tuple and display them.
- Q19.** Write a program to count the occurrence of each character in a given string using a dictionary.
- Q20.** Create a student record management system using dictionaries.
- Q21.** A bookstore wants to search whether a particular book exists in the inventory list.
- Q22.** Write a program to reverse each word in a given sentence.
- Q23.** A sports academy wants to display player names sorted alphabetically.
- Q24.** Create a program to merge two lists of employee IDs and remove duplicates.
- Q25.** Write a program to find common elements between two sets.
- Q26.** A shopping website wants to generate a receipt and store it in a text file.
- Q27.** Create a file handling program to count the number of words in a text file.
- Q28.** Write a program to append user feedback into a file without overwriting existing data.
- Q29.** A college wants to store attendance records and retrieve them using roll numbers.
- Q30.** Create a nested dictionary program to manage department-wise employee details.
- Q31.** A weather application wants to find the highest recorded temperature from weekly data.
- Q32.** Write a program to display multiplication tables using nested loops.
- Q33.** Create a billing system for a restaurant using functions and loops.
- Q34.** A courier company wants to track parcel IDs and their delivery status.
- Q35.** Write a program to validate whether a mobile number contains exactly 10 digits.
- Q36.** A hotel booking system wants to check room availability before booking.
- Q37.** Create a function that accepts a list of expenses and returns the highest expense.
- Q38.** Write a program to split a sentence into words and count total words.
- Q39.** A school wants to identify toppers from a list of student marks.
- Q40.** Create a password strength checker using string operations.
- Q41.** Write a program to calculate factorial using a while loop.
- Q42.** A company wants to store employee names and salaries, then display employees earning above a threshold.

- Q43.** Create a program to convert temperatures from Celsius to Fahrenheit for multiple values.
- Q44.** Write a function that returns the frequency of vowels in a paragraph.
- Q45.** A railway reservation system wants to verify whether a seat number exists in the reservation list.
- Q46.** Create a menu-driven calculator using functions.
- Q47.** Write a program to read names from a file and display them in sorted order.
- Q48.** A mobile recharge system wants to categorize recharge plans based on price ranges.
- Q49.** Create a program to generate usernames from employee full names.
- Q50.** Write a program to analyze sales data and identify the month with maximum sales.